

Amal Nath VJ

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Experience

UVJ Technology | Software Engineer (*Cold-chain logistics / Physics-informed ML*)

Trivandrum

- Engineered TempTrack AI, a proactive risk prediction platform for pharmaceutical cold-chain shipments, addressing \$50K+ losses from temperature excursions through pre-departure risk assessment and real-time monitoring.
- Architected dual-model ML system: Random Forest for pre-shipment evaluation (24hrs before departure) and XGBoost for in-transit monitoring, enabling logistics teams to prevent failures rather than react to them.
- Built physics-based thermal simulation engine modeling PCM phase transitions and heat transfer through vacuum-insulated containers, generating synthetic training data and physics-informed features for ML models.
- Developed automated QC report extraction pipeline using Claude API, reducing shipper onboarding from manual data entry to document-upload workflow while maintaining thermal modeling accuracy.

Siemens Technology and Services Pvt. Ltd | Intern

Chennai

- Worked on smart building projects involving access controllers (Sipass), SDT (System Design Tool), and surveillance systems.

Education

College of Engineering Trivandrum

2021 - 2025

B.Tech in Computer Science and Engineering

Projects

Credit Card Risk Prediction 🔒

Machine Learning, XGBoost, Pandas, Numpy

- Built an end-to-end credit card risk model to assess default risk on loan, customer, and bureau data, consolidating multiple sources and performing EDA + feature engineering.
- Designed, tuned, and compared Logistic Regression, Random Forest, and XGBoost models; addressed class imbalance and improved discrimination as measured by ROC-AUC, KS, and Gini.

Log Classification with Hybrid Framework 🔒

NLP, LLM, ML, FastAPI

- Built an intelligent log classification system for production system logs (errors, warnings, notifications) to reduce manual triage.
- Implemented a 3-tier hybrid pipeline (Regex, Sentence-Transformer + Logistic Regression, LLM with confidence-based fallback) to handle both structured and noisy legacy logs.
- Boosted classification accuracy by 38–42% and cut manual log triage effort by 30%.

Car Crash Damage Classification 🔒

DeepLearning, PyTorch, ResNet50

- Built a deep learning model to classify car damage from images into 6 categories (front/rear: breakage, crushed, normal) to support faster, consistent claim assessment.
- Fine-tuned a ResNet50 backbone with transfer learning and deployed a Streamlit app so users can upload images and get real-time damage predictions.

Flipkart E-Commerce Chatbot 🔒 Python, Streamlit, Groq, LLaMA 3.3, ChromaDB, Semantic Router, SQLite

- Built an end-to-end AI chatbot for an e-commerce platform that handles both FAQ queries and real-time product search using intent-based routing with Semantic Router.
- Implemented a RAG pipeline using ChromaDB and Sentence Transformers to retrieve relevant FAQ answers, and a Text-to-SQL pipeline using Groq (LLaMA 3.3) to query a SQLite product database based on natural language input.

Technical Skills

Languages: Python, SQL

ML / DL Frameworks: PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers, spaCy

NLP and LLMs: Sentence Transformers, LangChain, LLaMA, Groq API, Claude API, RAG, Semantic Router, ChromaDB

Backend Tech /Developer Tools: FastAPI , GitHub,Git

Domains: Machine Learning, Deep Learning , NLP , Large Language Models, GenAI , AgenticAI

Achievements & Leadership

Siemens Scholar (2021–2025): Selected among top engineering students in India based on academic merit.

Leader, CET Official Dance Crew (2021–2025): Led the team to the finals of the Destiny Dance Competition, representing Kerala at the national level.